

Introduction to software architecture for quantum computing systems special issue

A B S T R A C T

The rapid advancement of quantum computing is reshaping the landscape of software engineering, requiring new methods to software architecture. Unlike traditional software development, quantum software systems introduce unique challenges in design, implementation, and scalability, requiring well-defined architectural frameworks to ensure reliability and efficiency. This special issue presents research contributions that explore software architecture for quantum computing systems, addressing key aspects such as architectural decisions, quality assurance, scalability, and security. This editorial introduces the accepted papers and highlights their contributions to advancing the foundations of quantum software engineering.

1. Introduction

Quantum computing systems have emerged as a transformative technology, offering novel computational capabilities to address complex problem domains that surpass the limits of classical computing [1]. Major technology enterprises, including IBM, Google, and Microsoft, have heavily invested in the research and development of quantum systems, aiming to harness their potential for diverse industrial applications [2,3]. While significant advancements have been made in quantum hardware, the software engineering community has only recently begun focusing on establishing systematic techniques for quantum software development, including quantum software models, algorithmic specifications, and simulation evaluations.

Software architecture plays an important role in software-intensive systems by abstracting implementation-specific complexities into well-defined architectural components and connectors, ensuring structured system design and maintainability [4–6]. However, current quantum software development predominantly emphasizes quantum source code production while overlooking the importance of well-defined software architecture. This lack of architectural foundations results in systems that are prone to inefficiencies, defects, and scalability challenges [7].

The integration of software architecture principles into quantum software engineering is essential for organizing system requirements, guiding software design, and facilitating validation and deployment processes. Architectural frameworks and notations offer a structured model approach, to implement, and maintain quantum computing applications [8]. By empowering software practitioners with architectural methodologies tailored for quantum systems, it becomes possible to enhance software reliability, optimize resource utilization, and streamline development workflows.

This special issue aims to highlight cutting-edge research, methodologies, and practical insights on *Software Architecture for Quantum Computing Systems*. It provides a platform for researchers and practitioners to explore novel architectural paradigms, design principles, and

software engineering practices that can bridge the gap between quantum computing hardware and software-intensive applications. The contributions in this issue will cover topics such as architectural frameworks for quantum computing, best practices for quantum software design, and empirical studies on the impact of software architecture in quantum systems development.

2. Review process

We received 13 submissions in response to the call for papers. Before initiating the review process, an internal discussion among the associate editors was conducted to analyze the scope and novelty of each submission and to assign the most appropriate reviewers. One submission was desk rejected as it fell outside the scope of this special issue. The remaining 12 manuscripts were reviewed each by three reviewers. Following a thorough peer review process, including major and minor revisions, we ultimately accepted eight manuscripts for publication in this special issue.

3. Papers accepted in the special issue

In *QLSN: Quantum Key Distribution for Large Scale Networks* [9], Mangla et al. propose a novel quantum-inspired key management protocol designed to enhance security and reliability in large-scale networks. The study addresses key distribution challenges in networks susceptible to adversarial attacks, particularly in the era of quantum computing. The authors introduce QLSN, a protocol that replaces the classical Diffie–Hellman key exchange algorithm to mitigate risks associated with long-key transfers. The paper presents a risk analysis across different data sizes and evaluates the protocol's security and efficiency using IBM's Qiskit simulator. The results demonstrate that QLSN significantly improves network security and resilience, outperforming conventional IPsec-based solutions. The study highlights the potential of artificial intelligence and quantum computing fusion in revolutionizing

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future key management strategies .

In *Genetic Model-Based Success Probability Prediction of Quantum Software Development Projects* [10], Akbar et al. introduce an optimization-based predictive model for assessing the success probability of quantum software development (QSD) projects. The study identifies 14 critical variables influencing QSD success and employs Genetic Algorithms (GA) combined with Naïve Bayes and Logistic Regression models to predict project outcomes. The authors validate their model using expert survey data and demonstrate that as QSD processes mature, success probability increases while development costs decrease. The findings emphasize the importance of structured QSD methodologies, risk factor management, and cost-efficient project execution. The proposed model provides quantifiable insights for software practitioners to enhance decision-making in quantum software engineering .

In *Unraveling Quantum Computing System Architectures: An Extensive Survey of Cutting-Edge Paradigms* [11], Zhao et al. present a comprehensive survey on quantum software architectures, addressing the growing need for structured architectural frameworks in quantum computing. The study highlights how current quantum software development is primarily focused on source code generation while neglecting architectural considerations, which are critical for scalability, maintainability, and efficiency. By examining quantum software engineering practices, quantum modelling languages, quantum design patterns, and quantum communication security, the authors identify key architectural challenges and propose future directions for research. The findings emphasize the importance of architecture-driven quantum software development in ensuring robust and scalable quantum computing solutions .

Moreover in *"Role of Quantum Computing in Shaping the Future of 6G Technology* [12], Akbar, et al. explore the impact of quantum computing (QC) on the evolution of 6G wireless communication. The paper provides a dual-phase investigation, beginning with a literature review to identify 15 key QC applications in 6G and followed by an empirical study with industry experts to validate best practices for implementing QC in 6G networks. The study highlights how quantum technologies can enhance data security, optimize network performance, and enable advanced quantum machine learning for 6G applications. With a focus on practical implementation, the paper presents 49 best practices for integrating quantum computing into 6G infrastructures, offering valuable insights for both academic researchers and industry practitioners.

In *Quantum-Aided Efficient Resource Control for Connected Support in IRS-Assisted Networks* [13], Taneja et al. propose a quantum-enabled intelligent network framework for resource management in Intelligent Reflecting Surface (IRS)-assisted 6G networks. The study addresses the scarcity of radio spectrum and network resources, which limit the efficiency of future 6G systems. The authors introduce a quantum-based resource control algorithm that forms user clusters and assigns Access Point (AP)-IRS pairs, ensuring optimized beamforming and efficient network coverage. Their model is evaluated in terms of average sum rate, energy efficiency, and different clustering mechanisms, showing an improvement of 7.52% in the sum rate at 30dBm transmit power per AP. The results highlight the potential of quantum computing for network optimization, particularly in handling dynamic spectrum allocation and resource management in next-generation wireless networks .

In *Qubernetes: Towards a Unified Cloud-Native Execution Platform for Hybrid Classical-Quantum Computing* [14], Stirbu et al. introduce Qubernetes, an extension of Kubernetes designed to integrate quantum computing into cloud-native applications. The study addresses the challenge of orchestrating hybrid classical-quantum computations by leveraging Kubernetes' containerization capabilities. Using Design Science Research Methodology, the authors develop a quantum resource mapping framework that enables seamless scheduling and execution of quantum tasks alongside classical workloads. Their experimental results demonstrate that Qubernetes facilitates scalable and efficient execution of quantum algorithms by exposing quantum computation tasks through

standardized cloud-native principles. This work paves the way for leveraging Kubernetes as a foundational platform for hybrid quantum computing applications, bridging the gap between classical and quantum computing ecosystems.

In *Architecture Decisions in Quantum Software Systems: An Empirical Study on Stack Exchange and GitHub* [15], Shamima et al. investigate architecture decision-making in quantum software development by analyzing discussions from Stack Exchange and GitHub repositories. The study examines 385 GitHub issues from 87 quantum software projects and 70 Stack Exchange posts, categorizing architecture-related decisions into six linguistic patterns, with Solution Proposal and Information Giving being the most prevalent. The findings reveal that Implementation Decision and Technology Decisions are the dominant types of architectural choices, while Software Development Tools emerge as the most common application domain. The study also identifies maintainability as the most frequently considered quality attribute and highlights design issues and high error rates as major challenges in quantum software architecture. The research provides valuable insights into how developers approach architectural decision-making in quantum software systems and underscore the importance of structured architectural frameworks in advancing quantum software engineering.

Finally in *Service Engineering for Quantum Computing: Ensuring High-Quality Quantum Services* [16], Diaz et al. propose a quality assurance framework for quantum computing services, emphasizing service engineering principles to ensure the reliability, scalability, and maintainability of quantum applications. The authors present an extended OpenAPI Specification combined with SonarQube-based quality assessment tools, allowing for automated generation, deployment, and evaluation of quantum services. The study evaluates 40 implementations of quantum algorithms, revealing significant variability in analyzability across different quantum services. By integrating classical service engineering techniques with quantum computing principles, the proposed framework enhances quantum service quality and efficiency, paving the way for standardized quality metrics and best practices in quantum software development.

4. Conclusions

The articles in this Special Issue on Software Architecture for Quantum Computing Systems present diverse perspectives on architectural frameworks, quality assurance, scalability, and security in quantum software development. We hope that the insights provided in these articles will motivate and guide future research in this evolving field. Alongside the proposed architectural paradigms, these studies introduce novel methodologies and empirical findings that contribute to the systematic development of quantum software systems. Together, these contributions open up several research opportunities for advancing quantum software architecture, such as:

- Empirical validation of proposed quantum software architectures through real-world case studies to assess their applicability across different quantum computing platforms and domains.
- Comparative analysis of architectural frameworks and design patterns to identify the most effective approaches for optimizing quantum software performance, scalability, usability and maintainability.
- Development of standardized models, frameworks, and best practices that support the integration of quantum computing into mainstream software engineering methodologies.

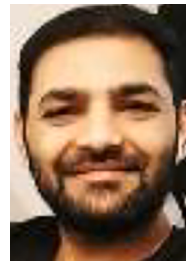
We believe that further research into architecture-driven approaches for quantum computing will significantly contribute to the growth of quantum software engineering, ensuring structured, scalable, and high-quality development practices for next-generation computing systems.

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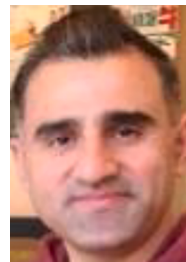
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